Exact Prescribed-Time Prescribed-Accuracy Control for Spacecraft under Dual Reaction Wheel Saturation

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I. Introduction

Increasingly complex spacecraft missions impose stringent requirements on both convergence time and accuracy of attitude control. Especially in time-critical missions such as rendezvous and docking, the spacecraft attitude must reach a target state at a prescribed terminal time T_f dictated by communication windows or multi-vehicle coordination [1–3]. Meanwhile, these missions also require the attitude to converge to within a prescribed accuracy imposed by payload pointing requirements [4–6]. For spacecraft actuated by reaction wheels, fulfilling these requirements is further complicated by dual saturation, namely, torque saturation and momentum saturation. Stringent terminal-time requirements drive high torques that accelerate momentum accumulation. Once momentum saturates, the spacecraft risks losing control authority.

Motivated by the demand for rapid convergence, finite-time control (FTC) and fixed-time control (FxTC) [7–10] have been extensively investigated, yet neither can prescribe T_f directly. Prescribed-time control (PTC) [11, 12] addresses this limitation through time-dependent scaling functions [13, 14] or time-varying high-gain feedback [15–17]. However, existing PTC strategies remain conservative. For instance, in Ref. [3], convergence occurs at 250 s for $T_f = 500$ s. In time-critical missions, such premature convergence may disrupt prescribed operation sequences and even lead to collisions, risking mission failure.

Beyond conservatism, the dual saturation constraint has been rarely addressed in existing PTC strategies, and its neglect entails two consequences. First, although torque saturation has been considered in certain PTC strategies [18–20], momentum saturation is seldom treated as a coupled constraint. Once momentum saturation occurs on an axis, the wheel cannot generate torque in that direction, compromising three-axis controllability. Second, T_f is typically treated as a free parameter without accounting for the minimum feasible maneuver time $T_{f,\min}$ imposed by

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dual saturation. Without $T_f \geq T_{f,\min}$, prescribed-time convergence is physically unrealizable, making the notion of prescribed time meaningless under dual saturation.

While dual saturation constraint can be explicitly incorporated through optimization-based trajectory planning [21–24], these methods are open-loop and therefore susceptible to unmodeled disturbances. Model predictive control (MPC) [25, 26] addresses disturbances through feedback, but the momentum saturation is a constraint of integral type, reducing the problem to a full-horizon optimization and leaving MPC ill-suited in this context. Approaches that combine trajectory planning with closed-loop tracking control [27, 28] offer robustness against perturbations, yet none provides a rigorous guarantee on the prescribed terminal time or tracking accuracy. Achieving exact prescribed-time convergence while rigorously accounting for dual-saturation constraints therefore remains an open problem.

To address this problem, this paper proposes an exact prescribed-time prescribed-accuracy control (EPTPAC) framework with a two-loop architecture. The outer loop solves a constrained optimal control problem (OCP) to generate a feasible reference trajectory that satisfies the dual-saturation constraints and reaches the target precisely at T_f. The inner loop employs a nonsingular prescribed-time controller that drives tracking errors to within a prescribed accuracy before T_f, despite inertia uncertainties and external disturbances. A parameter-synthesis method is developed in which most controller parameters are determined directly from mission inputs, with only three remaining as free design choices, coordinating the two loops to ensure closed-loop prescribed-accuracy performance.

The main contributions of this paper are as follows:

1) Exact convergence to within a prescribed accuracy at T_f is achieved through a two-loop architecture, despite inertia uncertainty and external disturbances.
2) Rigorous axis-by-axis feasibility conditions are derived to guarantee satisfaction of the dual-saturation constraint, and a minimum feasible maneuver time $T_{f,\min}$ is analytically quantified to ensure the prescribed terminal time is physically realizable.
3) A parameter-synthesis method is developed to coordinate the two-loop architecture with only three free design parameters, transforming the design process from heuristic to systematic.

II. Problem Description and Preliminaries

A. Problem Description

Let $\mathbb{R}$ denote the set of real numbers. $\|\cdot\|$ is the 2-norm of a vector or matrix, and $\boldsymbol{I}_n$ is the $n \times n$ identity matrix. $\lambda_{\min}(\boldsymbol{A})$ and $\lambda_{\max}(\boldsymbol{A})$ represent the extreme eigenvalues of $\boldsymbol{A}$. For $\boldsymbol{a} = [a_1, a_2, a_3]^\top \in \mathbb{R}^3$, $\boldsymbol{a}^\times$ denotes the skew-symmetric matrix $\boldsymbol{a}^\times \triangleq [0,\ -a_3,\ a_2;\ a_3,\ 0,\ -a_1;\ -a_2,\ a_1,\ 0]$.

The attitude kinematics and dynamics of a rigid spacecraft actuated by reaction wheels are governed by

$$\dot{\boldsymbol{\sigma}} = \boldsymbol{G}(\boldsymbol{\sigma})\boldsymbol{\omega}, \quad \boldsymbol{G}(\boldsymbol{\sigma}) = \frac{1}{4}\left[\left(1 - \|\boldsymbol{\sigma}\|^2\right)\boldsymbol{I}_3 + 2\boldsymbol{\sigma}^{\times} + 2\boldsymbol{\sigma}\boldsymbol{\sigma}^{\top}\right] \tag{1}$$

$$(\boldsymbol{J}_0 + \Delta\boldsymbol{J})\dot{\boldsymbol{\omega}} = -\boldsymbol{\omega}^{\times}[(\boldsymbol{J}_0 + \Delta\boldsymbol{J})\boldsymbol{\omega} + \boldsymbol{h}_w] + \boldsymbol{d} + \boldsymbol{\tau}, \quad \boldsymbol{\tau} = -\dot{\boldsymbol{h}}_w \tag{2}$$

where $\boldsymbol{\sigma} \in \mathbb{R}^3$ represents the attitude in Modified Rodrigues Parameters (MRPs), $\boldsymbol{\omega} \in \mathbb{R}^3$ denotes the body angular velocity, $\boldsymbol{J}_0 \in \mathbb{R}^{3\times3}$ is the nominal inertia with uncertainty $\Delta\boldsymbol{J}$, $\boldsymbol{J} \triangleq \boldsymbol{J}_0 + \Delta\boldsymbol{J}$ is the total inertia, $\boldsymbol{h}_w \in \mathbb{R}^3$ and $\boldsymbol{\tau} \in \mathbb{R}^3$ are the wheel momentum and control torque, respectively. $\boldsymbol{d} \in \mathbb{R}^3$ denotes the external disturbance.

A three-axis wheel cluster aligned with the body axes is subject to dual-saturation constraints

$$|\tau_i(t)| \leq \tau_{i,\max}, \quad |h_{w,i}(t)| \leq h_{w,i,\max}, \quad i \in \{1, 2, 3\} \tag{3}$$

where $\tau_{i,\max} > 0$ and $h_{w,i,\max} > 0$ are the torque and momentum saturation limits of the i-th axis, respectively.

Let $\boldsymbol{\sigma}_d, \boldsymbol{\omega}_d$ denote the desired attitude and angular velocity. The tracking errors $\boldsymbol{\sigma}_e, \boldsymbol{\omega}_e$ are defined as

$$\boldsymbol{\sigma}_e = \frac{(1 - \|\boldsymbol{\sigma}_d\|^2)\boldsymbol{\sigma} - (1 - \|\boldsymbol{\sigma}\|^2)\boldsymbol{\sigma}_d + 2\boldsymbol{\sigma}_d^{\times}\boldsymbol{\sigma}}{1 + \|\boldsymbol{\sigma}\|^2\|\boldsymbol{\sigma}_d\|^2 + 2\boldsymbol{\sigma}^{\top}\boldsymbol{\sigma}_d}, \qquad \boldsymbol{\omega}_e = \boldsymbol{\omega} - \boldsymbol{R}(\boldsymbol{\sigma}_e)\boldsymbol{\omega}_d \tag{4}$$

where $\boldsymbol{R}(\boldsymbol{\sigma}_e) = \boldsymbol{I}_3 + [\,8\boldsymbol{\sigma}_e{}^{\times}\boldsymbol{\sigma}_e{}^{\times} - 4(1 - \boldsymbol{\sigma}_e^{\top}\boldsymbol{\sigma}_e)\boldsymbol{\sigma}_e{}^{\times}\,]/(1 + \boldsymbol{\sigma}_e^{\top}\boldsymbol{\sigma}_e)^2$ is the direction cosine matrix. By the MRP shadow set property, $\|\boldsymbol{\sigma}_e\| < 1$ holds.

The error kinematics and dynamics take the form

$$\dot{\boldsymbol{\sigma}}_e = \boldsymbol{G}(\boldsymbol{\sigma}_e)\boldsymbol{\omega}_e, \quad \boldsymbol{G}(\boldsymbol{\sigma}_e) = \frac{1}{4}\left[\left(1 - \|\boldsymbol{\sigma}_e\|^2\right)\boldsymbol{I}_3 + 2\boldsymbol{\sigma}_e{}^{\times} + 2\boldsymbol{\sigma}_e\boldsymbol{\sigma}_e^{\top}\right] \tag{5}$$

$$\boldsymbol{J}_0\dot{\boldsymbol{\omega}}_e = \boldsymbol{\tau} + \boldsymbol{T}_d + \boldsymbol{f} \tag{6}$$

where $\boldsymbol{f} = -\boldsymbol{\omega}^{\times}(\boldsymbol{J}_0\boldsymbol{\omega} + \boldsymbol{h}_w) + \boldsymbol{J}_0\boldsymbol{\omega}_e^{\times}\boldsymbol{R}(\boldsymbol{\sigma}_e)\boldsymbol{\omega}_d - \boldsymbol{J}_0\boldsymbol{R}(\boldsymbol{\sigma}_e)\dot{\boldsymbol{\omega}}_d$ is the computable feedforward term, and $\boldsymbol{T}_d = -\Delta\boldsymbol{J}\dot{\boldsymbol{\omega}}_e - \boldsymbol{\omega}^{\times}\Delta\boldsymbol{J}\boldsymbol{\omega} + \Delta\boldsymbol{J}(\boldsymbol{\omega}_e^{\times}\boldsymbol{R}(\boldsymbol{\sigma}_e)\boldsymbol{\omega}_d - \boldsymbol{R}(\boldsymbol{\sigma}_e)\dot{\boldsymbol{\omega}}_d) + \boldsymbol{d}$ is the lumped perturbation. From $\|\boldsymbol{\sigma}_e\| < 1$, one obtains $\|\boldsymbol{G}(\boldsymbol{\sigma}_e)\| = (1 + \|\boldsymbol{\sigma}_e\|^2)/4 < \frac{1}{2}$.

The following assumptions are adopted throughout this note.

Assumption 1. *The prescribed terminal time satisfies*

$$T_f \geq T_{f,\min} \tag{7}$$

where $T_{f,\min} > 0$ is the minimum feasible maneuver time under the dual-saturation constraints (3). *A practical estimate* *of $T_{f,\min}$ follows from Appendix A.*

Assumption 2. *The lumped perturbation satisfies*

$$\|\boldsymbol{T}_d(t)\| \le T_{d,\max} \tag{8}$$

where $T_{d,\max} > 0$ is a known bound on the perturbation magnitude.

Assumption 3. *Let $\boldsymbol{H} = \boldsymbol{J}\omega + \boldsymbol{h}_w$ be the total momentum and $\boldsymbol{H}_0 = \boldsymbol{H}(0)$. The momentum drift induced by external disturbances satisfies*

$$\|\boldsymbol{H}(t) - \boldsymbol{H}_0\| \le H_{\max} \tag{9}$$

where $H_{\max} > 0$ is a known bound on the momentum drift.

B. Preliminaries

Consider the nonlinear system

$$\dot{\boldsymbol{x}} = \boldsymbol{g}(\boldsymbol{x}, t) \tag{10}$$

where $\boldsymbol{x} \in \mathbb{R}^n$ and $\boldsymbol{g}(\boldsymbol{x}, t) \in \mathbb{R}^n$ is locally Lipschitz in $\boldsymbol{x}$, with $\boldsymbol{g}(\boldsymbol{0}, t) \equiv \boldsymbol{0}$.

The origin of system (10) is said to be prescribed-time stable if there exists a prescribed constant $T_p > 0$ such that for any $\boldsymbol{x}_0 \in \mathbb{R}^n$ and all $t \ge T_p$, the solution $\boldsymbol{x}(t, \boldsymbol{x}_0)$ of system (10) satisfies $\boldsymbol{x}(t, \boldsymbol{x}_0) = \boldsymbol{0}$ [11]. The origin of system (10) is said to be prescribed-time prescribed-accuracy stable if there exist prescribed constants $T_p > 0$ and $\varepsilon > 0$ such that for any $\boldsymbol{x}_0 \in \mathbb{R}^n$ and all $t \ge T_p$, the solution $\boldsymbol{x}(t, \boldsymbol{x}_0)$ of system (10) satisfies $\|\boldsymbol{x}(t, \boldsymbol{x}_0)\| \le \varepsilon$ [29].

Lemma 1 ([30]). *Suppose there exists a Lyapunov function $V(\boldsymbol{x})$ for system* (10) *such that*

$$\dot{V} \le -\frac{\pi}{\eta T_p}\left(V^{1-\frac{\eta}{2}} + V^{1+\frac{\eta}{2}}\right) \tag{11}$$

where $0 < \eta < 1$ and $T_p > 0$ are given constants. Then the origin is prescribed-time stable.

To accommodate bounded perturbations, we extend Lemma 1 to the following practical prescribed-time prescribed-accuracy form.

Theorem 1. *Suppose there exists a Lyapunov function $V(\boldsymbol{x})$ for system* (10) *such that*

$$\dot{V} \le \delta V^{\frac{1}{2}} - \frac{\pi}{\eta T_p}\left[(1+\alpha)V^{1-\frac{\eta}{2}} + V^{1+\frac{\eta}{2}}\right] \tag{12}$$

where $\delta > 0$, $\alpha > 0$, $0 < \eta < 1$, and $T_p > 0$. Then the trajectory enters $V \le V_\delta^*$ with $V_\delta^* = \left(\frac{\delta\,\eta\,T_p}{\pi\,\alpha}\right)^{\frac{2}{1-\eta}}$ within T_p and remains therein.

Proof. Let $K = \frac{\pi}{\eta T_p}$. From (12), $\dot{V} \leq \delta V^{1/2} - K[(1+\alpha)V^{1-\frac{\eta}{2}} + V^{1+\frac{\eta}{2}}]$. At $V = V_\delta^*$, $\delta V^{1/2} = K\alpha V^{1-\eta/2}$ holds by definition. For all $V \geq V_\delta^*$, since $V^{-(1-\eta)/2}$ is monotonically decreasing, $\delta V^{1/2} \leq K\alpha V^{1-\eta/2}$, yielding $\dot{V} \leq -K(V^{1-\frac{\eta}{2}} + V^{1+\frac{\eta}{2}})$ for all $V \geq V_\delta^*$. By Lemma 1, any trajectory with $V(0) > V_\delta^*$ enters $\{V \leq V_\delta^*\}$ within T_p. At $V = V_\delta^*$, $\dot{V} \leq -K(V^{1-\frac{\eta}{2}} + V^{1+\frac{\eta}{2}})$ ensures forward invariance. □

III. Main Results

A. Outer-Loop Trajectory Planning Under Dual Saturation

A reference trajectory $(\boldsymbol{\sigma}_d(t), \boldsymbol{\omega}_d(t))$ is generated offline by solving the OCP over $[0, T_f]$, formulated as follows:

$$
\begin{aligned}
\min_{\boldsymbol{\tau}_{\text{ref}}(\cdot)} \quad & \mathcal{J} = \int_0^{T_f} \left(\|\boldsymbol{\tau}_{\text{ref}}(t)\|^2 + \lambda \|\dot{\boldsymbol{\tau}}_{\text{ref}}(t)\|^2 \right) dt \\
\text{s.t.} \quad & \dot{\boldsymbol{\sigma}}(t) = \boldsymbol{G}(\boldsymbol{\sigma}(t))\boldsymbol{\omega}(t), \\
& \boldsymbol{J}_0 \dot{\boldsymbol{\omega}}(t) + \boldsymbol{\omega}(t)^\times (\boldsymbol{J}_0 \boldsymbol{\omega}(t) + \boldsymbol{h}_w(t)) = \boldsymbol{\tau}_{\text{ref}}(t), \\
& \dot{\boldsymbol{h}}_w(t) = -\boldsymbol{\tau}_{\text{ref}}(t), \quad \boldsymbol{h}_w(0) = \boldsymbol{h}_{w,0}, \\
& \boldsymbol{\sigma}(0) = \boldsymbol{\sigma}_0, \quad \boldsymbol{\omega}(0) = \boldsymbol{\omega}_0, \\
& \boldsymbol{\sigma}(T_f) = \boldsymbol{\sigma}_{\text{target}}, \quad \boldsymbol{\omega}(T_f) = \boldsymbol{\omega}_{\text{target}}, \\
& |\tau_{\text{ref},i}(t)| \leq (1-\gamma_u)\tau_{i,\max}, \quad i \in \{1,2,3\}, \\
& |h_{w,i}(t)| \leq (1-\gamma_h) h_{w,i,\max}, \quad i \in \{1,2,3\}
\end{aligned}
\tag{13}
$$

where $\lambda > 0$ balances control effort against smoothness, and $\gamma_u, \gamma_h \in (0,1)$ are safety margins reserved for the inner loop. The OCP is solved via GPOPS-II [31], and the resulting discrete nodal solution is reconstructed as a continuous-time reference through piecewise-linear interpolation. This reference trajectory has bounded angular velocity and angular acceleration. Specifically, there exist constants $\omega_{d,\max} > 0$ and $\dot{\omega}_{d,\max} > 0$ such that $\|\boldsymbol{\omega}_d(t)\| \leq \omega_{d,\max}$ and $\|\dot{\boldsymbol{\omega}}_d(t)\| \leq \dot{\omega}_{d,\max}$ for all $t \in [0, T_f]$.

By construction, the outer loop provides a reference trajectory that reaches the target state exactly at T_f, thereby serving as the nominal reference for the inner-loop tracking controller.

B. Inner-Loop Nonsingular Prescribed-Time Prescribed-Accuracy Tracking Control

The sliding mode surface is designed as

$$
\boldsymbol{s} = \boldsymbol{\omega}_e + \boldsymbol{Q}(\boldsymbol{\sigma}_e) \tag{14}
$$

with the nonlinear coupling term $\boldsymbol{Q}(\boldsymbol{\sigma}_e)$ defined as

$$\boldsymbol{Q}(\boldsymbol{\sigma}_e) = c_1 k_1 \frac{\boldsymbol{\sigma}_e}{(\|\boldsymbol{\sigma}_e\| + \varepsilon_1)^{\eta}} + c_1 k_2 \frac{\boldsymbol{\sigma}_e}{(\|\boldsymbol{\sigma}_e\| + \varepsilon_1)^{-\eta}} \tag{15}$$

where $\varepsilon_1 > 0$, $T_{p1} > 0$, $\alpha_1 > 0$, and $\eta \in (0, 1)$ are design parameters. The regularization ε_1 in the denominator ensures that $\boldsymbol{Q}(\boldsymbol{\sigma}_e)$ remains nonsingular for all $\boldsymbol{\sigma}_e \in \mathbb{R}^3$. The gains are given by

$$c_1 = \frac{\pi}{\eta T_{p1}}, \quad k_1 = (1 + \alpha_1) 2^{1+\frac{3}{2}\eta}, \quad k_2 = 2^{1-\frac{\eta}{2}}$$

Theorem 2. *Consider the attitude error kinematics* (5) *with the sliding mode surface* (14)*, and suppose that the sliding mode variable $\boldsymbol{s}(t)$ remains within the bounded set $\|\boldsymbol{s}(t)\| \leq s_{\max}$. Then the attitude error $\boldsymbol{\sigma}_e(t)$ converges to the bounded set*

$$\|\boldsymbol{\sigma}_e(t)\| \leq \max\left\{\varepsilon_{\delta_1},\ \varepsilon_1\right\} \tag{16}$$

within T_{p1} and remains therein for all $t \geq T_{p1}$, where $\varepsilon_{\delta_1} = \sqrt{2}\,(\delta_1 \eta T_{p1}/(\pi \alpha_1))^{1/(1-\eta)}$ and $\delta_1 = s_{\max}/\sqrt{2}$.

Proof. Define the Lyapunov function candidate

$$V_1 = \frac{1}{2}\|\boldsymbol{\sigma}_e\|^2 \tag{17}$$

Differentiating V_1 along (5) and substituting $\boldsymbol{\omega}_e = \boldsymbol{s} - \boldsymbol{Q}(\boldsymbol{\sigma}_e)$ yields

$$\dot{V}_1 \leq \|\boldsymbol{G}(\boldsymbol{\sigma}_e)\|\,\|\boldsymbol{\sigma}_e\|\,s_{\max} - \boldsymbol{\sigma}_e^{\top}\boldsymbol{G}(\boldsymbol{\sigma}_e)\boldsymbol{Q}(\boldsymbol{\sigma}_e) \tag{18}$$

When $\|\boldsymbol{x}\| \geq \varepsilon$, the following norm inequalities hold:

$$\frac{\|\boldsymbol{x}\|^2}{(\|\boldsymbol{x}\| + \varepsilon)^{\eta}} \geq \frac{1}{2^{\eta}}\|\boldsymbol{x}\|^{2-\eta}, \qquad \|\boldsymbol{x}\|^2(\|\boldsymbol{x}\| + \varepsilon)^{\eta} \geq \|\boldsymbol{x}\|^{2+\eta} \tag{19}$$

For $\|\boldsymbol{\sigma}_e\| \geq \varepsilon_1$, applying these together with $\|\boldsymbol{G}(\boldsymbol{\sigma}_e)\| < \frac{1}{2}$ to $\dot{V}_1$ yields

$$\begin{aligned}\dot{V}_1 &\leq \frac{1}{2}\|\boldsymbol{\sigma}_e\| s_{\max} - \frac{c_1}{4}\left[k_1 \frac{\|\boldsymbol{\sigma}_e\|^2}{(\|\boldsymbol{\sigma}_e\| + \varepsilon_1)^{\eta}} + k_2 \frac{\|\boldsymbol{\sigma}_e\|^2}{(\|\boldsymbol{\sigma}_e\| + \varepsilon_1)^{-\eta}}\right] \\ &\leq \frac{\sqrt{2}}{2} V_1^{1/2} s_{\max} - \frac{c_1}{4}\left[k_1 2^{-\eta}(2V_1)^{1-\eta/2} + k_2 (2V_1)^{1+\eta/2}\right]\end{aligned} \tag{20}$$

Substituting the gains c_1, k_1, k_2, we obtain

$$\dot{V}_1 \le \delta_1 V_1^{1/2} - \frac{\pi}{\eta T_{p1}} \left[(1+\alpha_1) V_1^{1-\frac{\eta}{2}} + V_1^{1+\frac{\eta}{2}} \right] \tag{21}$$

This satisfies the condition of Theorem 1. Hence V_1 enters $\{V_1 < V_{\delta_1}^*\}$ before T_{p1} with $V_{\delta_1}^* = (\frac{\delta_1 \eta T_{p1}}{\pi \alpha_1})^{\frac{2}{1-\eta}}$. Expressing this bound in terms of $\|\boldsymbol{\sigma}_e\|$ yields $\|\boldsymbol{\sigma}_e(t)\| \le \sqrt{2V_{\delta_1}^*} = \varepsilon_{\delta_1}$. The regularized set $\{\|\boldsymbol{\sigma}_e\| \le \varepsilon_1\}$ is positively invariant. Thus $\|\boldsymbol{\sigma}_e(t)\| \le \max\{\varepsilon_{\delta_1}, \varepsilon_1\}$ for all $t \ge T_{p1}$. □

To drive the sliding variable $\boldsymbol{s}$ to a bounded neighborhood of the origin, the inner-loop control law is designed as

$$\boldsymbol{\tau} = -\boldsymbol{f}(t) - \boldsymbol{J}_0 \dot{\boldsymbol{Q}}(\boldsymbol{\sigma}_e) + \boldsymbol{\tau}_c \tag{22}$$

where

$$\boldsymbol{\tau}_c = -c_2 \left[k_3 \frac{\boldsymbol{s}}{(\|\boldsymbol{s}\| + \varepsilon_2)^{\eta}} + k_4 \frac{\boldsymbol{s}}{(\|\boldsymbol{s}\| + \varepsilon_2)^{-\eta}} \right] \tag{23}$$

The regularization $\varepsilon_2 > 0$ in the denominator ensures that $\boldsymbol{\tau}_c$ remains nonsingular for all $\boldsymbol{s} \in \mathbb{R}^3$. The gains are given by

$$c_2 = \frac{\pi}{\eta T_{p2}}, \quad k_3 = (1+\alpha_2) 2^{-1+\frac{3}{2}\eta} \lambda_{\max}(\boldsymbol{J}_0)^{1-\frac{\eta}{2}}, \quad k_4 = 2^{-1-\frac{\eta}{2}} \lambda_{\max}(\boldsymbol{J}_0)^{1+\frac{\eta}{2}} \tag{24}$$

where $T_{p2} > 0$ and $\alpha_2 > 0$ are design parameters. Substituting (22) into (6), together with the identity $\dot{\boldsymbol{\omega}}_e = \dot{\boldsymbol{s}} - \dot{\boldsymbol{Q}}(\boldsymbol{\sigma}_e)$, yields the closed-loop sliding dynamics

$$\boldsymbol{J}_0 \dot{\boldsymbol{s}} = \boldsymbol{\tau}_c + \boldsymbol{T}_d(t) \tag{25}$$

Theorem 3. *Consider the spacecraft tracking error system described by* (5) *and* (6)*, and suppose that Assumption 2 is satisfied. If the control law* (22) *is applied, then the sliding mode variable* $\boldsymbol{s}(t)$ *converges to the bounded set*

$$\|\boldsymbol{s}(t)\| \le \max\left\{\varepsilon_{\delta_2},\ \varepsilon_2\right\} \tag{26}$$

within T_{p2} *and remains therein for all* $t \ge T_{p2}$*, where* $\varepsilon_{\delta_2} = \sqrt{2/\lambda_{\min}(\boldsymbol{J}_0)}\, (\delta_2 \eta T_{p2}/(\pi\alpha_2))^{1/(1-\eta)}$ *and* $\delta_2 = T_{d,\max}\sqrt{2/\lambda_{\min}(\boldsymbol{J}_0)}$.

Proof. Define the Lyapunov function candidate

$$V_2 = \frac{1}{2} \boldsymbol{s}^\top \boldsymbol{J}_0 \boldsymbol{s} \tag{27}$$

Differentiating V_2 along (25) and using $\|\boldsymbol{s}\| \le \sqrt{2V_2/\lambda_{\min}(\boldsymbol{J}_0)}$, we obtain

$$\dot{V}_2 = \boldsymbol{s}^\top \boldsymbol{\tau}_c + \boldsymbol{s}^\top \boldsymbol{T}_d \le \boldsymbol{s}^\top \boldsymbol{\tau}_c + \delta_2 V_2^{1/2} \tag{28}$$

with $\delta_2 = T_{d,\max}\sqrt{2/\lambda_{\min}(\boldsymbol{J}_0)}$. For $\|\boldsymbol{s}\| \ge \varepsilon_2$, applying (19) and substituting the gains (24) yields

$$\begin{aligned} \boldsymbol{s}^\top \boldsymbol{\tau}_c &\le -c_2\Big[k_3 \frac{\|\boldsymbol{s}\|^2}{(\|\boldsymbol{s}\| + \varepsilon_2)^\eta} + k_4 \frac{\|\boldsymbol{s}\|^2}{(\|\boldsymbol{s}\| + \varepsilon_2)^{-\eta}}\Big] \\ &\le -c_2\big[k_3 2^{-\eta}\|\boldsymbol{s}\|^{2-\eta} + k_4\|\boldsymbol{s}\|^{2+\eta}\big] \\ &\le -\frac{\pi}{\eta T_{p2}}\big[(1+\alpha_2)V_2^{1-\eta/2} + V_2^{1+\eta/2}\big] \end{aligned} \tag{29}$$

Substituting this bound into the expression for $\dot{V}_2$ yields

$$\dot{V}_2 \le \delta_2 V_2^{1/2} - \frac{\pi}{\eta T_{p2}}\left[(1+\alpha_2)V_2^{1-\frac{\eta}{2}} + V_2^{1+\frac{\eta}{2}}\right] \tag{30}$$

This satisfies the condition of Theorem 1. Hence V_2 enters $\{V_2 \le V_{\delta_2}^*\}$ within T_{p2} with $V_{\delta_2}^* = (\frac{\delta_2 \eta T_{p2}}{\pi \alpha_2})^{\frac{2}{1-\eta}}$. Rewriting this bound in terms of $\|\boldsymbol{s}\|$ gives $\|\boldsymbol{s}(t)\| \le \sqrt{2V_{\delta_2}^*/\lambda_{\min}(\boldsymbol{J}_0)} = \varepsilon_{\delta_2}$. The set $\{\|\boldsymbol{s}\| \le \varepsilon_2\}$ is positively invariant. Thus $\|\boldsymbol{s}(t)\| \le \max\{\varepsilon_{\delta_2}, \varepsilon_2\}$ for all $t \ge T_{p2}$. □

Theorems 2 and 3 jointly ensure two-stage convergence. Theorem 3 guarantees $\|\boldsymbol{s}(t)\| \le \max\{\varepsilon_{\delta_2}, \varepsilon_2\}$ for all $t \ge T_{p2}$, satisfying $\|\boldsymbol{s}\| \le s_{\max}$ required by Theorem 2. The attitude error then converges to $\|\boldsymbol{\sigma}_e(t)\| \le \max\{\varepsilon_{\delta_1}, \varepsilon_1\}$ within T_{p1} thereafter. By appropriately selecting $\{T_{p1}, T_{p2}\}$ and $\{\alpha_1, \alpha_2\}$, one can ensure $T_{p1} + T_{p2} < T_f$, $\varepsilon_{\delta_1} < \varepsilon_1$, and $\varepsilon_{\delta_2} < \varepsilon_2$, guaranteeing convergence to the prescribed accuracy within the prescribed time T_f.

The following theorems verify dual-saturation compliance under the condition

$$\|\boldsymbol{\sigma}_e(t)\| \le \varepsilon_1, \quad \|\boldsymbol{s}(t)\| \le \varepsilon_2 \tag{31}$$

To bound the torque deviation caused by tracking errors and inertia uncertainty, we decompose the feedforward term $-\boldsymbol{f}$ in the control law (22) as $-\boldsymbol{f} = \boldsymbol{\tau}_{\mathrm{ref}} + \boldsymbol{\Delta}_f$, where $\boldsymbol{\tau}_{\mathrm{ref}}$ is the reference torque and $\boldsymbol{\Delta}_f$ captures the mismatch. Under the disturbance-free dynamics of the OCP, the total reference momentum $\boldsymbol{H}_d = \boldsymbol{J}_0\boldsymbol{\omega}_d + \boldsymbol{h}_{w,d}$ is conserved. Hence, with $\boldsymbol{H}_d(0) = \boldsymbol{H}(0) = \boldsymbol{H}_0$ and $H_0 \triangleq \|\boldsymbol{H}_0\|$, we have

$$\|\boldsymbol{H}_d(t)\| = H_0, \qquad \|\boldsymbol{H}_d(t) - \boldsymbol{H}_0\| \le 2H_0. \tag{32}$$

The reference torque is $\boldsymbol{\tau}_{\text{ref}} = \boldsymbol{J}_0\dot{\boldsymbol{\omega}}_d + \boldsymbol{\omega}_d^{\times}\boldsymbol{H}_d$, and substituting $\boldsymbol{f}$ from (6) into $\boldsymbol{\Delta}_f = -\boldsymbol{f} - \boldsymbol{\tau}_{\text{ref}}$ yields

$$\begin{aligned}\boldsymbol{\Delta}_f &= \boldsymbol{\omega}^{\times}(\boldsymbol{H} - \boldsymbol{H}_0) + \boldsymbol{\omega}_e^{\times}\boldsymbol{H}_0 + [(\boldsymbol{R} - \boldsymbol{I})\boldsymbol{\omega}_d]^{\times}\boldsymbol{H}_0 + \boldsymbol{\omega}_d^{\times}(\boldsymbol{H}_0 - \boldsymbol{H}_d) \\ &\quad - \boldsymbol{\omega}^{\times}\Delta\boldsymbol{J}\boldsymbol{\omega} - \boldsymbol{J}_0\boldsymbol{\omega}_e^{\times}\boldsymbol{R}\boldsymbol{\omega}_d + \boldsymbol{J}_0(\boldsymbol{R} - \boldsymbol{I})\dot{\boldsymbol{\omega}}_d. \end{aligned} \tag{33}$$

Invoking $\|\boldsymbol{H} - \boldsymbol{H}_0\| \leq H_{\max}$ from Assumption 3, $\|\boldsymbol{H}_0 - \boldsymbol{H}_d\| \leq 2H_0$ from (32), $\|\boldsymbol{I} - \boldsymbol{R}\| \leq 4\|\boldsymbol{\sigma}_e\|$, and $\|\boldsymbol{a}^{\times}\boldsymbol{b}\| \leq \|\boldsymbol{a}\|\|\boldsymbol{b}\|$, the component-wise bound is

$$\begin{aligned}|\Delta_{f,i}| &\leq (\|\boldsymbol{\omega}_e\| + \|\boldsymbol{\omega}_d\|)H_{\max} + \|\boldsymbol{\omega}_e\|H_0 + 4\|\boldsymbol{\sigma}_e\|\|\boldsymbol{\omega}_d\|H_0 + 2\|\boldsymbol{\omega}_d\|H_0 \\ &\quad + \hat{\nu}_i(\|\boldsymbol{\omega}_e\| + \|\boldsymbol{\omega}_d\|)^2 + \mu_i\|\boldsymbol{\omega}_e\|\|\boldsymbol{\omega}_d\| + 4\mu_i\|\boldsymbol{\sigma}_e\|\|\dot{\boldsymbol{\omega}}_d\| \end{aligned} \tag{34}$$

where $\mu_i = \sum_{j=1}^{3}|(J_0)_{ij}|$, $\nu_i = \sum_{j=1}^{3}|(\Delta J)_{ij}|$, $\hat{\nu}_i = \sum_{j\neq i}\nu_j$.

Theorem 4. *Consider the inner-loop control law* (22)*. Under the condition* (31)*, if the safety margin γ_u and prescribed-accuracy set $(\varepsilon_1, \varepsilon_2)$ are selected such that*

$$\Delta_{f,i,\max}(\varepsilon_1, \varepsilon_2) + \overline{(J_0\dot{Q})_i}(\varepsilon_1, \varepsilon_2) + \bar{\tau}_c(\varepsilon_2) \leq \gamma_u\,\tau_{i,\max}, \qquad i = 1, 2, 3 \tag{35}$$

where the bounds are defined in the proof below, then the actuator torques satisfy $|\tau_i(t)| \leq \tau_{i,\max}$ for all $i = 1, 2, 3$.

Proof. Substituting the decomposition $-\boldsymbol{f} = \boldsymbol{\tau}_{\text{ref}} + \boldsymbol{\Delta}_f$ and the control law (22) into the expression for $\boldsymbol{\tau}$, for each axis $i \in \{1, 2, 3\}$, the triangle inequality yields

$$|\tau_i| \leq |\tau_{\text{ref},i}| + |\Delta_{f,i}| + |(\boldsymbol{J}_0\dot{\boldsymbol{Q}})_i| + |\tau_{c,i}| \tag{36}$$

The outer loop ensures $|\tau_{\text{ref},i}(t)| \leq (1 - \gamma_u)\tau_{i,\max}$ for all $t \in [0, T_f]$. Under condition (31), the angular velocity error satisfies

$$\|\boldsymbol{\omega}_e(t)\| \leq \varepsilon_2 + \bar{Q} \tag{37}$$

where

$$\bar{Q} = \sup_{\|\boldsymbol{\sigma}_e\| \leq \varepsilon_1} \|\boldsymbol{Q}(\boldsymbol{\sigma}_e)\| \leq c_1\left(k_1 2^{-\eta}\varepsilon_1^{1-\eta} + k_2 2^{\eta}\varepsilon_1^{1+\eta}\right) \tag{38}$$

(i) Bound on $|\Delta_{f,i}|$. Substituting the bounds (38) and the reference bounds from Section III into the feedforward

mismatch decomposition (34), for each $i = 1, 2, 3$, yields

$$|\Delta_{f,i}(t)| \leq \Delta_{f,i,\max}(\varepsilon_1, \varepsilon_2) = (\varepsilon_2 + \bar{Q} + \omega_{d,\max})H_{\max} + (\varepsilon_2 + \bar{Q})H_0 + 4\varepsilon_1\omega_{d,\max}H_0 + 2\omega_{d,\max}H_0 + \hat{\nu}_i(\varepsilon_2 + \bar{Q} + \omega_{d,\max})^2 + \mu_i(\varepsilon_2 + \bar{Q})\omega_{d,\max} + 4\mu_i\varepsilon_1\dot{\omega}_{d,\max} \tag{39}$$

where μ_i, ν_i, $\hat{\nu}_i$ are as defined in (34), and $H_{\max}$ is from Assumption 3.

(ii) Bound on $|\tau_{c,i}|$. Since $\|\boldsymbol{s}(t)\| \leq \varepsilon_2$, for each $i = 1, 2, 3$, we have

$$|\tau_{c,i}(t)| \leq \bar{\tau}_c(\varepsilon_2) = c_2\big(k_3 2^{-\eta}\varepsilon_2^{1-\eta} + k_4 2^{\eta}\varepsilon_2^{1+\eta}\big) \tag{40}$$

(iii) Bound on $|(\boldsymbol{J}_0\dot{\boldsymbol{Q}})_i|$. Differentiating $\boldsymbol{Q}$ yields $\dot{\boldsymbol{Q}} = \frac{\partial \boldsymbol{Q}}{\partial \boldsymbol{\sigma}_e}\boldsymbol{G}(\boldsymbol{\sigma}_e)\boldsymbol{\omega}_e$. Since $\|\boldsymbol{G}(\boldsymbol{\sigma}_e)\| = (1+\|\boldsymbol{\sigma}_e\|^2)/4 \leq (1+\varepsilon_1^2)/4$ for $\|\boldsymbol{\sigma}_e\| \leq \varepsilon_1$,

$$\|\dot{\boldsymbol{Q}}(t)\| \leq \left\|\frac{\partial \boldsymbol{Q}}{\partial \boldsymbol{\sigma}_e}\right\| \frac{1+\varepsilon_1^2}{4}\,(\varepsilon_2 + \bar{Q}) \tag{41}$$

For $\|\boldsymbol{\sigma}_e\| \leq \varepsilon_1$, the Jacobian norm is bounded by

$$\left\|\frac{\partial \boldsymbol{Q}}{\partial \boldsymbol{\sigma}_e}\right\| \leq c_1\Big[k_1(2\varepsilon_1)^{-\eta} + k_2(2\varepsilon_1)^{\eta} + \varepsilon_1\big(k_1\eta(2\varepsilon_1)^{-\eta-1} + k_2\eta(2\varepsilon_1)^{\eta-1}\big)\Big] \triangleq \Psi(\varepsilon_1) \tag{42}$$

Hence

$$|(\boldsymbol{J}_0\dot{\boldsymbol{Q}})_i| \leq \mu_i\,\Psi(\varepsilon_1)\,\frac{1+\varepsilon_1^2}{4}\,(\varepsilon_2 + \bar{Q}) \triangleq \overline{(J_0\dot{Q})}_i(\varepsilon_1, \varepsilon_2) \tag{43}$$

Finally, substituting the bounds (i)–(iii) into (36), for each axis i, the control torque satisfies

$$|\tau_i(t)| \leq (1-\gamma_u)\tau_{i,\max} + \Delta_{f,i,\max}(\varepsilon_1, \varepsilon_2) + \overline{(J_0\dot{Q})}_i(\varepsilon_1, \varepsilon_2) + \bar{\tau}_c(\varepsilon_2)$$

Consequently, if condition (35) holds, the control torque satisfies $|\tau_i(t)| \leq \tau_{i,\max}$ for all $i \in \{1, 2, 3\}$. □

Theorem 5. *Consider the inner-loop control law* (22)*. Under the condition* (31)*, if the safety margin* γ_h *and prescribed-accuracy set* $(\varepsilon_1, \varepsilon_2)$ *are selected such that*

$$H_{\max} + 2H_0 + 4\mu_i\varepsilon_1\omega_{d,\max} + (\mu_i + \nu_i)(\varepsilon_2 + \bar{Q}) + \nu_i\omega_{d,\max} \leq \gamma_h\, h_{w,i,\max}, \qquad i = 1, 2, 3 \tag{44}$$

where $\mu_i = \sum_{j=1}^{3}|(J_0)_{ij}|$ *and* $\nu_i = \sum_{j=1}^{3}|(\Delta J)_{ij}|$*, then the wheel momentum* $|h_{w,i}(t)| \leq h_{w,i,\max}$ *holds for all* $i = 1, 2, 3$.

Proof. The wheel momentum $\boldsymbol{h}_w = \boldsymbol{H} - (\boldsymbol{J}_0 + \Delta\boldsymbol{J})\boldsymbol{\omega}$ can be related to the reference momentum $\boldsymbol{h}_{w,d} = \boldsymbol{H}_d(t) - \boldsymbol{J}_0\boldsymbol{\omega}_d$

as follows. Using $\boldsymbol{\omega} = \boldsymbol{\omega}_e + \boldsymbol{R}(\boldsymbol{\sigma}_e)\boldsymbol{\omega}_d$ and writing $\boldsymbol{H} = \boldsymbol{H}_0 + (\boldsymbol{H} - \boldsymbol{H}_0)$ and $\boldsymbol{H}_d = \boldsymbol{H}_0 + (\boldsymbol{H}_d - \boldsymbol{H}_0)$, we obtain

$$\boldsymbol{h}_w = \boldsymbol{h}_{w,d} + (\boldsymbol{H} - \boldsymbol{H}_0) + (\boldsymbol{H}_0 - \boldsymbol{H}_d) - \boldsymbol{J}_0\big(\boldsymbol{R}(\boldsymbol{\sigma}_e) - \boldsymbol{I}\big)\boldsymbol{\omega}_d - (\boldsymbol{J}_0 + \Delta\boldsymbol{J})\boldsymbol{\omega}_e - \Delta\boldsymbol{J}\boldsymbol{R}(\boldsymbol{\sigma}_e)\boldsymbol{\omega}_d$$

Under condition (31), the error satisfies $\|\boldsymbol{\omega}_e\| \le \varepsilon_2 + \bar{Q}$ and $\|\boldsymbol{\sigma}_e\| \le \varepsilon_1$. The reference trajectory satisfies $\|\boldsymbol{\omega}_d\| \le \omega_{d,\max}$ from Section III and $|h_{w,d,i}| \le (1-\gamma_h)h_{w,i,\max}$. From Assumption 3, $\|\boldsymbol{H} - \boldsymbol{H}_0\| < H_{\max}$, and inequality (32) gives $\|\boldsymbol{H}_0 - \boldsymbol{H}_d(t)\| \le 2H_0$. Additionally, $\|\boldsymbol{I} - \boldsymbol{R}(\boldsymbol{\sigma}_e)\| \le 4\|\boldsymbol{\sigma}_e\| \le 4\varepsilon_1$. Applying the triangle inequality to each axis i then yields

$$\begin{aligned}
|h_{w,i}| &\le |h_{w,d,i}| + H_{\max} + 2H_0 + 4\mu_i\varepsilon_1\omega_{d,\max} + (\mu_i + \nu_i)\|\boldsymbol{\omega}_e\| + \nu_i\|\boldsymbol{\omega}_d\| \\
&\le |h_{w,d,i}| + H_{\max} + 2H_0 + 4\mu_i\varepsilon_1\omega_{d,\max} + (\mu_i + \nu_i)(\varepsilon_2 + \bar{Q}) + \nu_i\omega_{d,\max} \\
&\le (1-\gamma_h)h_{w,i,\max} + H_{\max} + 2H_0 + 4\mu_i\varepsilon_1\omega_{d,\max} + (\mu_i + \nu_i)(\varepsilon_2 + \bar{Q}) + \nu_i\omega_{d,\max}
\end{aligned} \tag{45}$$

Therefore, if (44) holds, the wheel momentum satisfies $|h_{w,i}(t)| \le h_{w,i,\max}$ for all $i = 1, 2, 3$. □

Remark 1. *With $\boldsymbol{\sigma}_d(0) = \boldsymbol{\sigma}(0)$ and $\boldsymbol{\omega}_d(0) = \boldsymbol{\omega}(0)$, the initial errors are zero. By Theorems 2 and 3, the sets $\{\|\boldsymbol{s}\| \le \varepsilon_2\}$ and $\{\|\boldsymbol{\sigma}_e\| \le \varepsilon_1\}$ are forward invariant. Hence the conditions of Theorems 4 and 5 hold over $[0, T_f]$, ensuring dual-saturation compliance throughout the maneuver.*

Remark 2. *Within the prescribed accuracy sets, $\boldsymbol{\Delta}_f$ and $\boldsymbol{J}_0\dot{\boldsymbol{\omega}}_e$ remain relatively small, so the torque margin is predominantly dictated by $\boldsymbol{T}_d$, giving the practical guideline $\gamma_u > T_{d,\max}/\tau_{i,\max}$ for each axis i. Similarly, the term $(\mu_i + \nu_i)(\varepsilon_2 + \bar{Q})$ in the momentum bound remains relatively small, so the momentum margin is governed by $H_{\max}$, $2H_0$, and $\nu_i\omega_{d,\max}$, giving $\gamma_h > (H_{\max} + 2H_0 + \nu_i\omega_{d,\max})/h_{w,i,\max}$ for each axis i.*

C. Parameter-Synthesis Method

The parameter-synthesis procedure is as follows.

Step 1. Mission inputs. Specify T_f, $\varepsilon_{\text{mission}}$, $\tau_{i,\max}$, $h_{w,i,\max}$, $T_{d,\max}$, and $H_{\max}$.

Step 2. Margins and feasibility. Select γ_u, γ_h via Remark 2 and verify $T_f \ge T_{f,\min}$ via Appendix A.

Step 3. Inner-loop synthesis. Set $\varepsilon_1 = \varepsilon_{\text{mission}}$, $\eta \in (0,1)$, and $T_{p1}, T_{p2} > 0$ with $T_{p1} + T_{p2} < T_f$. Select $\varepsilon_2 > 0$ such that the required gains satisfy

$$\alpha_1 \ge \frac{\delta_1 \eta T_{p1}}{\pi}\left(\frac{\sqrt{2}}{\varepsilon_1}\right)^{1-\eta}, \qquad \alpha_2 \ge \frac{\delta_2 \eta T_{p2}}{\pi}\left(\frac{\sqrt{2}}{\varepsilon_2\sqrt{\lambda_{\min}(\boldsymbol{J}_0)}}\right)^{1-\eta} \tag{46}$$

where $s_{\max} = \varepsilon_2$, $\delta_1 = s_{\max}/\sqrt{2}$, and $\delta_2 = T_{d,\max}\sqrt{2/\lambda_{\min}(\boldsymbol{J}_0)}$.

Step 4. Saturation verification. Check conditions in Theorems 4–5. If violated, increase T_f, relax ε_1, or increase γ_u, γ_h.

For a given mission, the framework requires only three free designer choices, namely η, T_{p1}, and T_{p2}, with all remaining parameters systematically derived via the analytical conditions, thus systematizing the design process. The complete procedure is illustrated in Fig. 1.

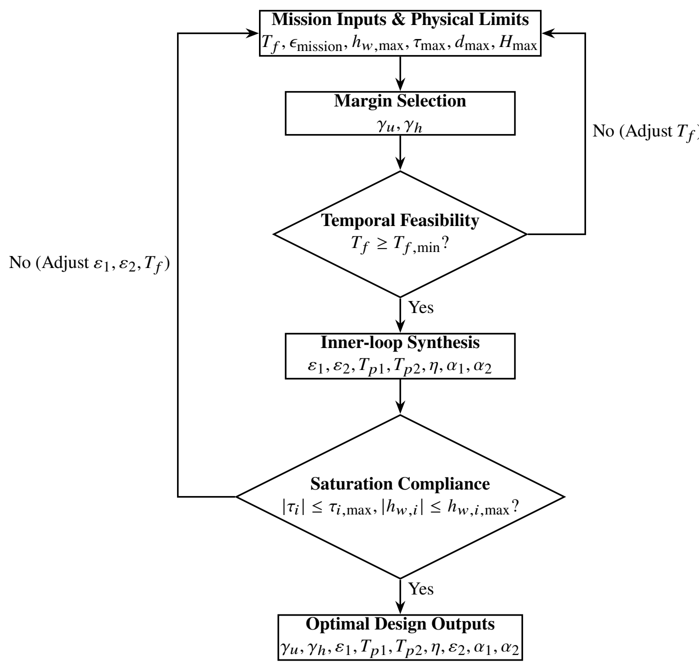


Fig. 1 Parameter-synthesis method.

Remark 3. *To avoid unbalanced gains, enforce* $\alpha_1 = \alpha_2 = \kappa\,\alpha$ *with* $\kappa > 1$ *by solving* $\alpha_{1,\min} = \alpha_{2,\min}$ *in* (46). *This yields*

$$\varepsilon_2 = \left(\frac{2T_{d,\max}\,T_{p2}}{T_{p1}\lambda_{\min}(\boldsymbol{J}_0)^{1-\frac{\eta}{2}}} \right)^{\frac{1}{2-\eta}} \varepsilon_1^{\frac{1-\eta}{2-\eta}}, \quad \alpha = \frac{\delta_1\,\eta\,T_{p1}}{\pi} \left(\frac{\sqrt{2}}{\varepsilon_1} \right)^{1-\eta} \tag{47}$$

followed by $\alpha_1 = \alpha_2 = \kappa\,\alpha$. *This provides a unique parameter set from mission inputs.*

IV. Simulation Results

This section presents simulation results to validate the proposed EPTPAC framework. For all simulations, the initial attitude is set to $\boldsymbol{\sigma}(0) = [0.2,\ 0.3,\ -0.3]^\top$ and the target attitude is $\boldsymbol{\sigma}_{\text{target}} = \mathbf{0}$. The spacecraft starts and ends at rest with $\boldsymbol{\omega}(0) = \boldsymbol{\omega}_{\text{target}} = \mathbf{0}$ rad/s. The common physical and disturbance parameters are listed in Table 1.

In each case, the nominal reference trajectory $(\boldsymbol{\sigma}_d, \boldsymbol{\omega}_d)$ is generated offline by solving the constrained OCP (13) via GPOPS-II, and is held constant at the target state for $t \geq T_f$. Following Remark 2, $\gamma_u = 0.2$ and $\gamma_h = 0.1$ are adopted.

Table 1 Common simulation settings

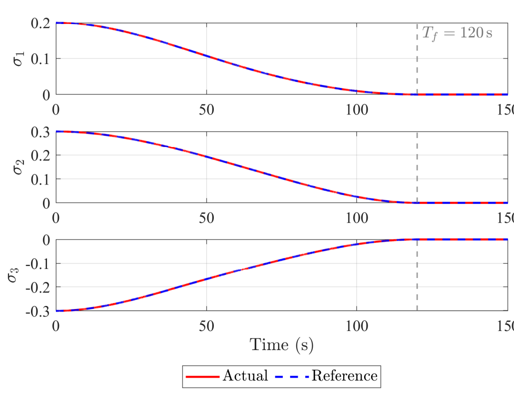
Symbol	Value
$\boldsymbol{J}_0$	$\mathrm{diag}([200,\ 250,\ 300])\ \mathrm{kg}\cdot\mathrm{m}^2$
$\Delta\boldsymbol{J}$	$0.05\boldsymbol{J}_0$
$\tau_{i,\max}$	$0.2\,\mathrm{N}\cdot\mathrm{m}$
$h_{w,i,\max}$	$4.0\,\mathrm{kg}\cdot\mathrm{m}^2/\mathrm{s}$
$\boldsymbol{h}_{w,0}$	$[0,\ 0,\ 0]^\top\ \mathrm{kg}\cdot\mathrm{m}^2/\mathrm{s}$
$\boldsymbol{d}(t)$	$0.01\,[\sin(2t),\ \cos(t),\ \cos(t+2)]^\top\ \mathrm{N}\cdot\mathrm{m}$
$T_{d,\max}$	$0.03\,\mathrm{N}\cdot\mathrm{m}$
$H_{\max}$	$0.1\,\mathrm{kg}\cdot\mathrm{m}^2/\mathrm{s}$

From Appendix A, the minimum feasible terminal time is $T_{f,\min} \approx 95$ s. Therefore, $T_f \geq 110$ s is employed in all cases to ensure feasibility.

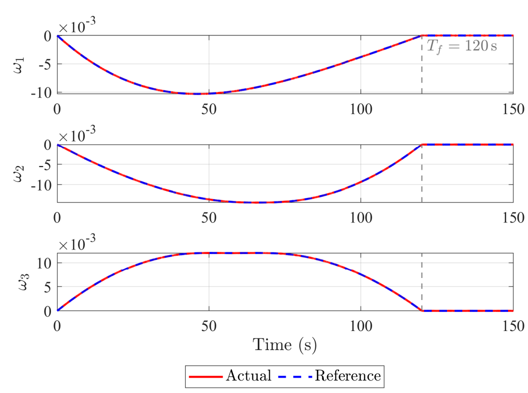
To comprehensively evaluate the framework, three cases are conducted: Case 1 validates the baseline performance; Case 2 assesses exact prescribed-time convergence and accuracy tunability; and Case 3 benchmarks the proposed framework against the torque-saturated prescribed-time controller in [19] (hereafter Ref. [19]) under both torque-only and dual-saturation conditions.

A. Case 1: Nominal Performance Validation

The mission parameters are taken as follows: terminal time $T_f = 120$ s, mission accuracy $\varepsilon_1 = 10^{-5}$, prescribed-time exponent $\eta = 0.2$, and time allocations $T_{p2} = 15$ s, $T_{p1} = 100$ s. Following the synthesis method in Section III.C with $\kappa = 1.2$, the inner-loop parameters are determined to be $\varepsilon_2 = 6.43 \times 10^{-5}$ and $\alpha_1 = \alpha_2 = 6.34$. The results are as follows.



(a) Attitude MRPs



(b) Angular velocity

Fig. 2 Case 1: Reference tracking in MRPs and angular rates.

Fig. 2 shows that the spacecraft attitude and angular velocity closely track the reference trajectory, which reaches the target state at $t = T_f$. Furthermore, Fig. 3 shows that all tracking errors remain within the prescribed-accuracy sets over $[0, T_f]$, while Fig. 4 confirms that both the control torque and wheel momentum remain within their respective saturation limits throughout the maneuver.

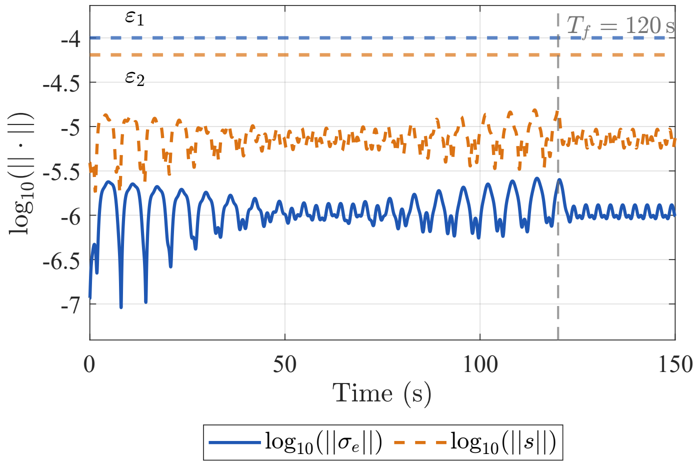
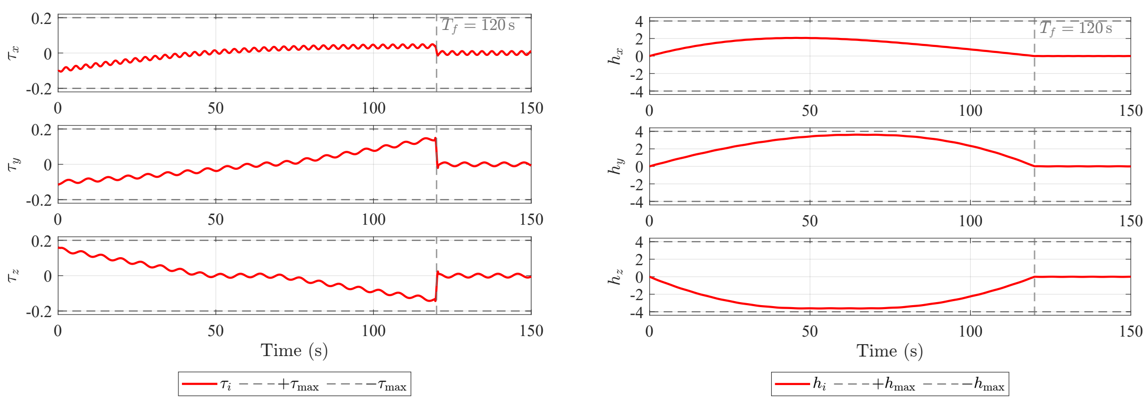


Fig. 3 Case 1: Log-scale error norms and prescribed-accuracy bounds.



(a) $\boldsymbol{\tau}$

(b) $\boldsymbol{h}_w$

Fig. 4 Case 1: Torque and wheel momentum.

In summary, Case 1 confirms that the EPTPAC framework achieves exact prescribed-time convergence to the target state while satisfying the dual-saturation constraint.

B. Case 2: Prescribed Performance Assessment

This case evaluates the prescribed-time convergence and prescribed-accuracy tunability of the EPTPAC framework.

1. Prescribed-time convergence.

The prescribed terminal time is swept over $T_f \in \{110,\ 120,\ 130,\ 140\}$ s, with $\varepsilon_1 = 10^{-5}$, $\eta = 0.2$, $T_{p2} = 15$ s, and $T_{p1} = T_f - 20$ s. For each T_f, the inner-loop parameters are synthesized via the balanced rule with $\kappa = 1.2$ following the procedure in Section III.C. The results are as follows.

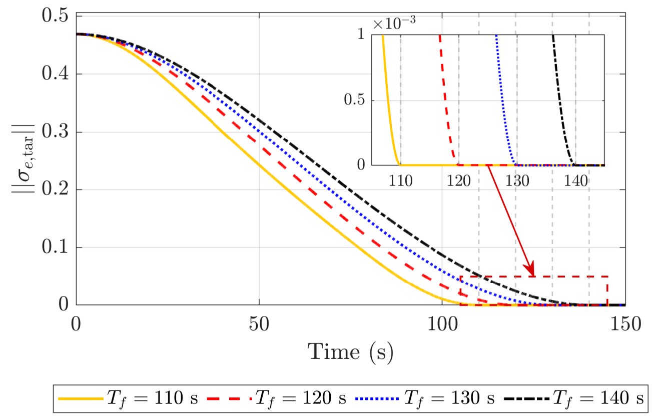
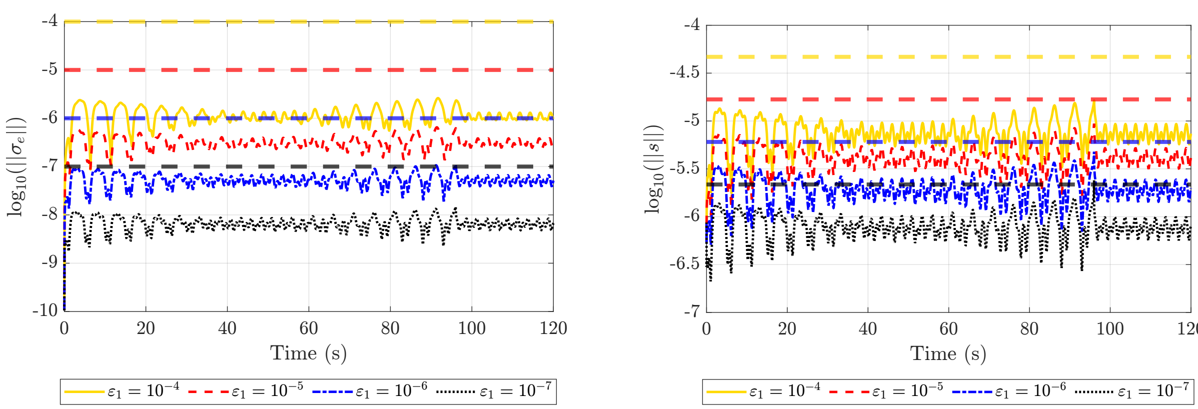


Fig. 5 Case 2: MRP error norm for different T_f

Fig. 5 shows that the attitude tracking error converges at the respective prescribed terminal times for all tested values, entering the prescribed accuracy only 0.24–0.27 s prior to T_f.

2. Prescribed-accuracy tunability.

To evaluate prescribed-accuracy tunability, ε_1 is swept over $\{10^{-4}, 10^{-5}, 10^{-6}, 10^{-7}\}$ with $T_f = 120\,\mathrm{s}$ fixed. For each ε_1, the inner-loop parameters are synthesized via the balanced rule with $\kappa = 1.2$ following Section III.C.



(a) $\log_{10}(\|\boldsymbol{\sigma}_e\|)$ with ε_1 thresholds

(b) $\log_{10}(\|\boldsymbol{s}\|)$ with ε_2 thresholds

Fig. 6 Case 2: Log-scale comparison of $\|\sigma_e\|$ and $\|s\|$ for different ε_1

Fig. 6 shows that all four settings achieve their prescribed accuracies within T_f. However, as the prescribed accuracy tightens, the required gains grow substantially: reducing ε_1 from 10^{-4} to 10^{-7} increases $\alpha_1 = \alpha_2$ from 4.30 to 24.78. Thus ε_1 should be matched to the actual sensor resolution, since pursuing accuracy beyond the sensing capability merely escalates control gains without practical benefit.

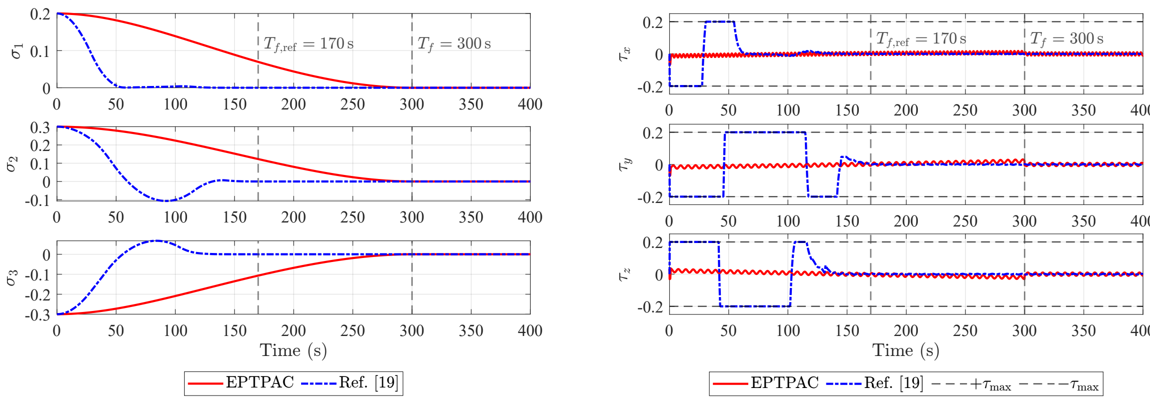
In summary, Case 2 validates the exact prescribed-time convergence and prescribed-accuracy tunability through sweeps of T_f and ε_1, respectively.

C. Case 3: Comparison with Ref. [19]

To evaluate the advantage of the proposed framework over existing methods, this case compares EPTPAC with Ref. [19], which addresses torque limits but does not account for momentum constraints. For fairness, all physical parameters, external disturbances, and initial conditions are kept identical to those in Table 1. Two scenarios are considered: Case 3.1 evaluates torque saturation only under identical constraint assumptions to provide a fair comparison, while Case 3.2 enforces dual saturation to assess how the momentum constraint affects controller performance.

1. Case 3.1: Torque-only saturation

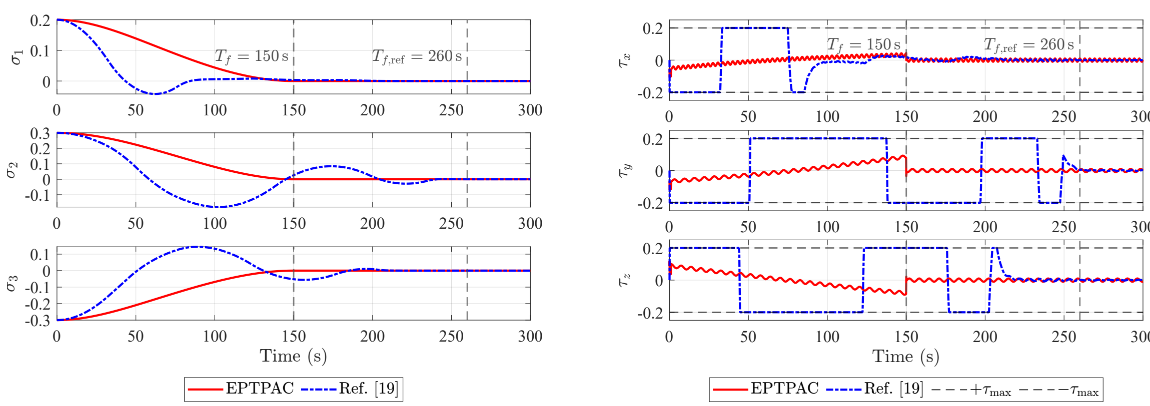
This scenario evaluates prescribed-time convergence under torque-only saturation ($\tau_{i,\max} = 0.2$ N · m) with two prescribed terminal times: $T_f = 300$ s and $T_f = 150$ s. The comparison results are as follows.



(a) Attitude error (b) Control torque

Fig. 7 Case 3.1: Torque-only saturation with $T_f = 300$ s

Fig. 7 shows that for $T_f = 300$ s, Ref. [19] converges around 150 s, exhibiting pronounced conservatism at only 50% of the prescribed maneuver duration.



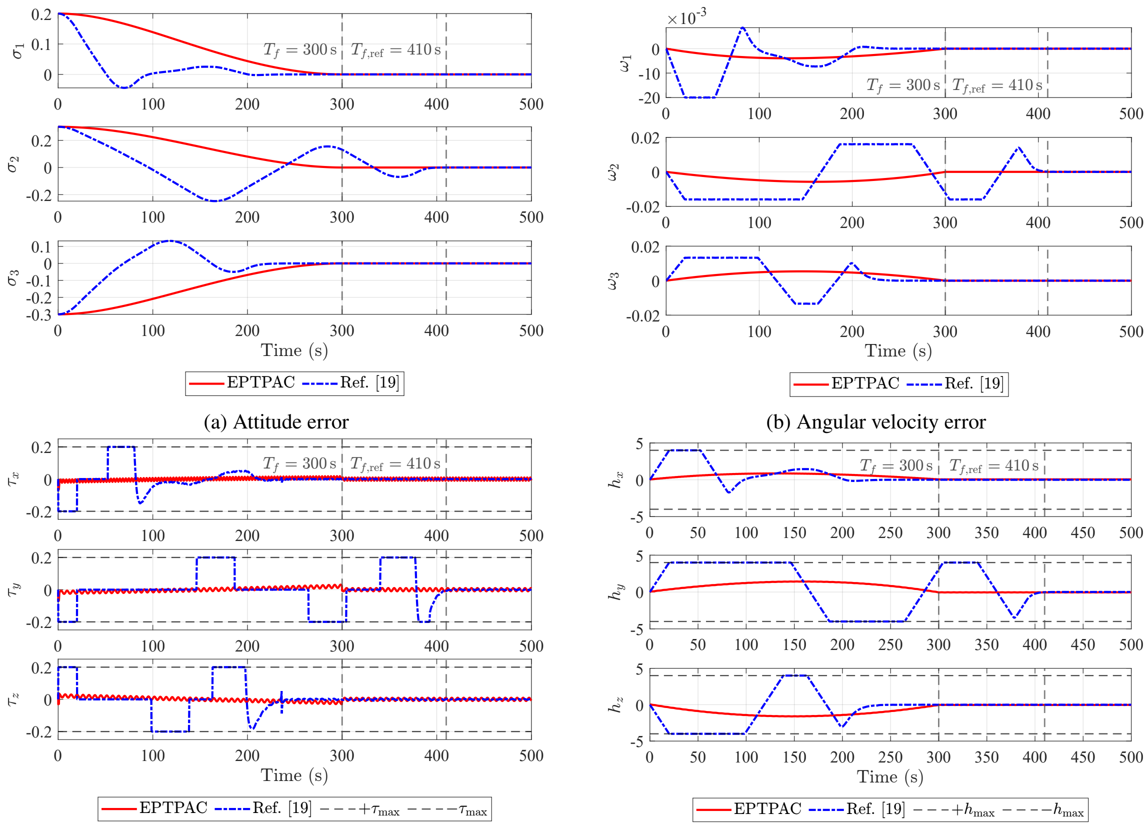
(a) Attitude error (b) Control torque

Fig. 8 Case 3.1: Torque-only saturation with $T_f = 150$ s

However, when T_f is reduced to 150 s, the prescribed-time convergence no longer holds. Fig. 8 shows that the response does not converge at 150 s but is instead delayed to approximately 230 s. This phenomenon stems from the aggressive escalation of time-varying gains as T_f approaches $T_{f,\min}$, which drives the control torque into saturation and invalidates the theoretical conditions on which convergence depends. In contrast, EPTPAC achieves exact convergence at both prescribed terminal times with smooth, feasible control torques.

2. *Case 3.2: Dual saturation*

This scenario examines controller performance under dual-saturation constraints with $T_f = 300$ s.



(a) Attitude error

(b) Angular velocity error

(c) Control torque

(d) Wheel momentum

Fig. 9 Case 3.2: Performance comparison under dual saturation

As shown in Fig. 9, under dual saturation Ref. [19] fails to achieve prescribed-time convergence, with the settling time delayed to approximately 410 s. This failure stems from the lack of momentum management: aggressive torque commands rapidly drive the wheels into saturation, leading to loss of control authority and uncontrolled drift. In contrast, EPTPAC modulates the control torques to prevent momentum saturation, achieving exact convergence at $t = 300$ s.

Through comparative simulations, Case 3 demonstrates the advantage of EPTPAC over Ref. [19]. Specifically, EPTPAC eliminates the conservatism inherent in conventional PTC methods and achieves exact prescribed-time convergence. Moreover, it preserves maneuver feasibility under dual-saturation constraints throughout the maneuver.

V. Conclusion

This paper addresses the problem of exact prescribed-time attitude control under dual reaction wheel saturation, which poses two principal challenges. First, existing PTC methods are inherently conservative, guaranteeing only an upper bound on convergence time rather than convergence exactly at a prescribed instant. Second, dual saturation severely constrains maneuver feasibility, where ignoring momentum saturation risks loss of control authority and neglecting $T_{f,\min}$ may render prescribed-time convergence physically unattainable. The proposed EPTPAC framework addresses these challenges through a decoupled two-loop architecture, eliminating conservatism while ensuring dual-saturation compliance. Moreover, an analytical estimate of $T_{f,\min}$ guarantees physical realizability of the prescribed terminal time, and a parameter-synthesis method further makes the design process systematic rather than heuristic. Numerical simulations validated the effectiveness of the proposed framework in achieving exact terminal-time convergence, tunable accuracy, and dual-saturation compliance.

A. Estimate of Minimum Feasible Maneuver Time

This appendix provides a practical estimate of $T_{f,\min}$ for use in Assumption 1. Given the initial attitude error $\boldsymbol{\sigma}_e(0)$, the principal rotation angle θ and axis $\boldsymbol{e}$ satisfy $\theta = 4\arctan(\|\boldsymbol{\sigma}_e(0)\|)$ and $\boldsymbol{e} = \boldsymbol{\sigma}_e(0)/\|\boldsymbol{\sigma}_e(0)\|$ (or any unit vector if $\boldsymbol{\sigma}_e(0) = \mathbf{0}$). Define $J_e = \boldsymbol{e}^\top \boldsymbol{J}_0 \boldsymbol{e}$. Under component-wise torque limits $|\tau_i| \le \tau_{i,\max}$, the maximum torque projection along $\boldsymbol{e}$ is $\tau_{e,\max} = \sum_{i=1}^{3} |e_i| \tau_{i,\max}$.

Assuming $\boldsymbol{h}_w(0) = \mathbf{0}$ and a bang-bang allocation $\tau_i = \mathrm{sgn}(e_i)\tau_{i,\max}$, the maximum duration before momentum saturation is $t_h = \min_i h_{w,i,\max}/\tau_{i,\max}$. For a rest-to-rest rotation about $\boldsymbol{e}$, the peak rate is $\omega_{\max} = \tau_{e,\max} t_h / J_e$ and the no-coast angle is $\theta_h = \tau_{e,\max} t_h^2 / J_e$. Then, the minimum-time estimate is

$$T_{f,\min} \approx \begin{cases} 2\sqrt{J_e \theta / \tau_{e,\max}}, & \theta \le \theta_h, \\ 2t_h + (\theta - \theta_h)/\omega_{\max}, & \theta > \theta_h \end{cases} \tag{48}$$

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References

[1] Alex Pothen, A., Crain, A., and Ulrich, S., "Pose Tracking Control for Spacecraft Proximity Operations Using the Udwadia–Kalaba Framework," *Journal of Guidance, Control, and Dynamics*, Vol. 45, No. 2, 2022, pp. 296–309. https://doi.org/10.2514/1.G005169, URL https://doi.org/10.2514/1.G005169, publisher: American Institute of Aeronautics and Astronautics _eprint: https://doi.org/10.2514/1.G005169.

[2] Das, G., and Sinha, M., "Predefined-Time Hybrid Tracking Control of Flexible Satellites with Conditional Anti-Unwinding," *Journal of Guidance, Control, and Dynamics*, Vol. 47, No. 11, 2024, pp. 2426–2434. https://doi.org/10.2514/1.G008286, URL https://doi.org/10.2514/1.G008286.

[3] Xu, C., Zelazo, D., and Wu, B., "Distributed prescribed-time coordinated control of spacecraft formation flying under input saturation," *Advances in Space Research*, Vol. 74, No. 5, 2024, pp. 2302–2315. https://doi.org/10.1016/j.asr.2024.05.077, URL https://www.sciencedirect.com/science/article/pii/S027311772400543X.

[4] Xiao, Y., J. de Ruiter, A. H., Ye, D., and Sun, Z., "Attitude Tracking Control for Rigid-Flexible Coupled Spacecraft with Guaranteed Performance Bounds," *Journal of Guidance, Control, and Dynamics*, Vol. 43, No. 2, 2020, pp. 327–337. https://doi.org/10.2514/1.G004535.

[5] Ye, D., Zou, A.-M., and Sun, Z., "Predefined-Time Predefined-Bounded Attitude Tracking Control for Rigid Spacecraft," *IEEE Transactions on Aerospace and Electronic Systems*, Vol. 58, No. 1, 2022, pp. 464–472. https://doi.org/10.1109/TAES.2021.3103258, URL https://ieeexplore.ieee.org/abstract/document/9512481.

[6] Xiao, Y., Yang, Y., Ye, D., and Zhang, J., "Quantitative Precision Second-Order Temporal Transformation-Based Pose Control for Spacecraft Proximity Operations," *IEEE Transactions on Aerospace and Electronic Systems*, Vol. 61, No. 2, 2025, pp. 1931–1941. https://doi.org/10.1109/TAES.2024.3469167.

[7] Hong, Y., and Jiang, Z.-P., "Finite-Time Stabilization of Nonlinear Systems With Parametric and Dynamic Uncertainties," *IEEE Transactions on Automatic Control*, Vol. 51, No. 12, 2006, pp. 1950–1956. https://doi.org/10.1109/TAC.2006.886515, URL https://ieeexplore.ieee.org/document/4026648/.

[8] Hu, Q., Tan, X., and Akella, M. R., "Finite-Time Fault-Tolerant Spacecraft Attitude Control with Torque Saturation," *Journal of Guidance, Control, and Dynamics*, Vol. 40, No. 10, 2017, pp. 2524–2537. https://doi.org/10.2514/1.G002191.

[9] Polyakov, A., "Nonlinear Feedback Design for Fixed-Time Stabilization of Linear Control Systems," *IEEE Transactions on Automatic Control*, Vol. 57, No. 8, 2012, pp. 2106–2110. https://doi.org/10.1109/TAC.2011.2179869, URL https://ieeexplore.ieee.org/document/6104367/.

[10] Polyakov, A., Efimov, D., and Perruquetti, W., "Finite-time and fixed-time stabilization: Implicit Lyapunov function approach," *Automatica*, Vol. 51, 2015, pp. 332–340. https://doi.org/10.1016/j.automatica.2014.10.082, URL https://www.sciencedirect.com/science/article/pii/S0005109814004634.

[11] Sánchez-Torres, J. D., Sanchez, E. N., and Loukianov, A. G., "Predefined-time stability of dynamical systems with sliding modes," *2015 American Control Conference (ACC)*, 2015, pp. 5842–5846. https://doi.org/10.1109/ACC.2015.7172255, URL https://ieeexplore.ieee.org/document/7172255, iSSN: 2378-5861.

[12] Song, Y., Wang, Y., Holloway, J., and Krstic, M., "Time-varying feedback for regulation of normal-form nonlinear systems in prescribed finite time," *AUTOMATICA*, Vol. 83, 2017, pp. 243–251. https://doi.org/10.1016/j.automatica.2017.06.008, URL https://linkinghub.elsevier.com/retrieve/pii/S000510981730290X, num Pages: 9 Place: Oxford Publisher: Pergamon-Elsevier Science Ltd Web of Science ID: WOS:000408288800028.

[13] Sánchez-Torres, J. D., Gómez-Gutiérrez, D., López, E., and Loukianov, A. G., "A class of predefined-time stable dynamical systems," *IMA Journal of Mathematical Control and Information*, Vol. 35, No. Supplement_1, 2018, pp. i1–i29. https://doi.org/10.1093/imamci/dnx004, URL https://doi.org/10.1093/imamci/dnx004.

[14] Xiao, Y., Wang, Y., Ye, D., and Sun, Z., "Predefined-time robust attitude tracking control of flexible spacecraft with continuous and nonsingular performance," *Aerospace Science and Technology*, Vol. 159, 2025, p. 110000. https://doi.org/10.1016/j.ast.2025.110000, URL https://www.sciencedirect.com/science/article/pii/S1270963825000720.

[15] Zhou, B., "Finite-time stability analysis and stabilization by bounded linear time-varying feedback," *Automatica*, Vol. 121, 2020, p. 109191. https://doi.org/10.1016/j.automatica.2020.109191, URL https://www.sciencedirect.com/science/article/pii/S0005109820303897.

[16] Cao, Y., Cao, J., and Song, Y., "Practical prescribed time tracking control over infinite time interval involving mismatched uncertainties and non-vanishing disturbances," *Automatica*, Vol. 136, 2022, p. 110050. https://doi.org/10.1016/j.automatica.2021.110050, URL https://www.sciencedirect.com/science/article/pii/S0005109821005781.

[17] Xiao, Y., Yang, Y., Ye, D., and Zhao, Y., "Scaling-Transformation-Based Attitude Tracking Control for Rigid Spacecraft With Prescribed Time and Prescribed Bound," *IEEE Transactions on Aerospace and Electronic Systems*, Vol. 61, No. 1, 2025, pp. 433–442. https://doi.org/10.1109/TAES.2024.3451454.

[18] Sun, H.-J., Wu, Y.-Y., and Zhang, J., "Attitude synchronization control for multiple spacecraft: A preassigned finite-time scheme," *Advances in Space Research*, Vol. 73, No. 12, 2024, pp. 6094–6110. https://doi.org/10.1016/j.asr.2024.02.001, URL https://www.sciencedirect.com/science/article/pii/S0273117724001303.

[19] Xu, C., Wu, B., and Wang, D., "Distributed Prescribed-Time Attitude Coordination for Multiple Spacecraft With Actuator Saturation Under Directed Graph," *IEEE Transactions on Aerospace and Electronic Systems*, Vol. 58, No. 4, 2022, pp. 2660–2672. https://doi.org/10.1109/TAES.2021.3135484, URL https://ieeexplore.ieee.org/document/9650527/.

[20] Zhao, S., Zheng, J., Yi, F., Wang, X., and Zuo, Z., "Exponential Predefined Time Trajectory Tracking Control for Fixed-Wing UAV With Input Saturation," *IEEE Transactions on Aerospace and Electronic Systems*, Vol. 60, No. 5, 2024, pp. 6406–6419. https://doi.org/10.1109/TAES.2024.3402656, URL https://ieeexplore.ieee.org/document/10534850/.

[21] Kobayashi, H., Shoji, Y., and Yamada, K., "Optimal Attitude Trajectory of Spacecraft with Pyramid-Type CMGs Using a Pseudo-Spectral Method," *AEROSPACE TECHNOLOGY JAPAN, THE JAPAN SOCIETY FOR AERONAUTICAL AND SPACE SCIENCES*, Vol. 16, No. 0, 2017, pp. 55–63. https://doi.org/10.2322/astj.16.55, URL https://www.jstage.jst.go.jp/article/astj/16/0/16_JSASS-D-16-00034/_article/-char/ja/.

[22] Spiller, D., Melton, R. G., and Curti, F., "Inverse dynamics particle swarm optimization applied to constrained minimum-time maneuvers using reaction wheels," *Aerospace Science and Technology*, Vol. 75, 2018, pp. 1–12. https://doi.org/10.1016/j.ast.2017.12.038, URL https://www.sciencedirect.com/science/article/pii/S1270963817308684.

[23] Celani, F., and Lucarelli, D., "Spacecraft Attitude Motion Planning Using Gradient-Based Optimization," *Journal of Guidance, Control, and Dynamics*, Vol. 43, No. 1, 2020, pp. 140–145. https://doi.org/10.2514/1.G004531, URL https://arc.aiaa.org/doi/10.2514/1.G004531, publisher: American Institute of Aeronautics and Astronautics.

[24] Fan, Y., Xu, R., Li, Z., Zhu, S., Shang, H., Gao, A., and Gao, Y., "Polynomial Attitude Trajectory Planning for Spacecraft with Movable Parts Using Decoupled Strategy," *Journal of Guidance, Control, and Dynamics*, Vol. 47, No. 1, 2024, pp. 72–86. https://doi.org/10.2514/1.G007645, URL https://doi.org/10.2514/1.G007645.

[25] Mammarella, M., Lee, D. Y., Park, H., Capello, E., and Ghiglino, P., "Attitude Control of a Small Spacecraft via Tube-Based Model Predictive Control," *Journal of Spacecraft and Rockets*, Vol. 56, No. 6, 2019, pp. 1585–1597. https://doi.org/10.2514/1.A34394.

[26] Yang, Y., "Attitude Model Predictive Control with Actuator Saturation Using an Arc-Search Interior-Point Method," *Journal of Guidance, Control, and Dynamics*, Vol. 46, No. 5, 2023, pp. 969–978. https://doi.org/10.2514/1.G007179.

[27] Biggs, J. D., and Colley, L., "Geometric Attitude Motion Planning for Spacecraft with Pointing and Actuator Constraints," *Journal of Guidance, Control, and Dynamics*, Vol. 39, No. 7, 2016, pp. 1672–1677. https://doi.org/10.2514/1.G001514.

[28] Yue, C., Huo, T., Lu, M., Shen, Q., Li, C., Chen, X., and Cao, X., "A Systematic Method for Constrained Attitude Control Under Input Saturation," *IEEE Transactions on Aerospace and Electronic Systems*, Vol. 59, No. 5, 2023, pp. 6005–6015. https://doi.org/10.1109/TAES.2023.3268607.

[29] Xiao, Y., Yang, Y., Ye, D., and Zhao, Y., "Scaling-Transformation-Based Attitude Tracking Control for Rigid Spacecraft With Prescribed Time and Prescribed Bound," *IEEE Transactions on Aerospace and Electronic Systems*, Vol. 61, No. 1, 2025, pp. 433–442. https://doi.org/10.1109/TAES.2024.3451454, URL https://ieeexplore.ieee.org/document/10659135.

[30] Anguiano-Gijón, C. A., Muñoz-Vázquez, A. J., Sánchez-Torres, J. D., Romero-Galván, G., and Martínez-Reyes, F., "On predefined-time synchronisation of chaotic systems," *Chaos, Solitons & Fractals*, Vol. 122, 2019, pp. 172–178. https://doi.org/10.1016/j.chaos.2019.03.015, URL https://www.sciencedirect.com/science/article/pii/S0960077919300864.

[31] Patterson, M. A., and Rao, A. V., "GPOPS-II: A MATLAB Software for Solving Multiple-Phase Optimal Control Problems Using hp-Adaptive Gaussian Quadrature Collocation Methods and Sparse Nonlinear Programming," *ACM Trans. Math. Softw.*, Vol. 41, No. 1, 2014, pp. 1:1–1:37. https://doi.org/10.1145/2558904, URL https://dl.acm.org/doi/10.1145/2558904.